USF Bellini College Senior Project

The Bellini College of AI, Cybersecurity and Computing at the University of South Florida is seeking industry participation in its Senior Project course (CIS 4910) for **Spring 2026**. Projects should entail software and/or hardware development. Teams of three to five students will complete the projects in one semester. Final results will be presented to the industry mentor at the semester end. The faculty member responsible for this course is:

Scot Hollingsworth
Adjunct Faculty
Bellini College of AI, Cybersecurity and Computing
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If you are interested in submitting a project, please fill-in this form and return it to Ken Christensen. Good projects are technical in nature, slightly open-ended, not on the critical path, and not proprietary.

Name: _Redwan Alqasemi_____ **Title:** _Research Professor_____

Email: _alqasemi@usf.edu_____ **Phone:** _(813)974-2115_____

Company name: _Center for Assistive, Rehabilitation and Robotics Technologies (CARRT)_____

Willing to give a presentation on a technical topic (please check one): Yes ( ) No (X)

Project area (please check one): Software ( ) Hardware ( ) Software and hardware (X)

Project idea: Robotic Arm Manipulation for Performing Activities of Daily Living

Description: Individuals with diminished physical capabilities must often rely on assistants to perform Activities of Daily Living (ADL). However, these individuals are usually in possession of sound cognitive and perceptual abilities. The goal of this project is to create a human-robot collaborative system that takes advantage of the impaired individual's cognitive abilities to control a robotic arm with superior dexterity, range of motion and grasping power to complete simple tasks within a defined environment.

As part of this project, the existing Kinova Gen 3 robotic arm with 7 degrees of freedom (DoF) will be used in combination with the onboard RGBD camera (Intel RealSense D435) to perform at least 3 simple-but-meaningful tasks related to ADL. These meaningful tasks can be selected by the team working on this project and may include tasks such as "making a bowl of cereal" or "putting objects inside a drawer". The Robot's base will be stationary and the workspace for the selected tasks should be within the range of the robot. While the system should be able to recognize the change in the location of objects, the environment and objects themselves can be assumed to be invariable. The user should be able to manipulate the robot through a tablet interface and voice commands. The user should also have the ability to interrupt any task and provide corrective action when needed (as part of the human-in-the-loop control system)

Desired Milestones:

1) Implement an object recognition algorithm capable of recognizing key objects for pre-defined tasks and identifying their location on the image.
2) Develop an algorithm combining the data from the previous step, the camera location and the depth map, captured by the camera, to find the location of objects relative to the robot.
3) Implement an algorithm to manipulate the robot to go to the desired location and perform primitive actions, such as "Grasping", "Placing", "Dropping" and "Pouring". Collectively, these primitive actions should lead to a complete ADL task.

4) Create a user interface, on a tablet, where the user can select tasks, override the actions taken by robot, see the camera feed, manually select objects and primitive actions.
5) Implement voice commands as part of the UI.
6) Test the system and collect qualitative and quantitative data to measure the effectiveness of the system in performing the ADL tasks.
7) Write a comprehensive report and prepare a paper for publication.

Note: Students are required to submit a timeline for their project with clear milestones by the second week of classes and a progress report weekly thereafter. Access to the lab and the needed equipment will be offered to the students working on the project for the whole semester.